Dear Hiring Team,

I am applying for the Analyst, Data Science role in the Model Risk Management Group. I recently completed my M.Sc. in Computer Science at Sapienza University of Rome, and the work I have done maps closely to what this role asks for.

Model testing and evaluation. My thesis evaluated pretrained Vision Transformers under continual learning. I benchmarked adapter-based methods and Elastic Weight Consolidation against ResNet-18 and naive fine-tuning baselines, measuring accuracy, forgetting, and computational efficiency, and validated results across seeds. The final model reached 90.59% average accuracy with 2.01% average forgetting on Split CIFAR-100.

Generative AI and LLM systems. I built and deployed Knowledge Copilot, a RAG chatbot using the Gemini API and Pinecone. Every answer is grounded in retrieved source text and returned with citations to limit hallucination. I split the pipeline into four independently testable stages, so I could trace a failure to chunking, retrieval, or generation. This taught me to look at where an LLM system can fail, not only at whether it works.

Analytical checks and monitoring. In my metal defect detection project, I compared a CNN against SVM and Random Forest models using confusion matrices, precision, recall, and F1, rather than accuracy alone. In DocDelta, I deployed on AWS with CloudWatch monitoring for latency and errors. My IEEE paper compared several ML models on accuracy, robustness, and generalization.

Communication. I have documented research findings with my thesis group, presented a peer-reviewed paper, and written project documentation that explains design choices and results clearly.

I should be upfront that I have not worked in a formal model risk or banking environment, and my hands-on tools are Python and SQL rather than PySpark or R. I am keen to learn regulatory and governance frameworks for AI, and I am comfortable reading research and challenging assumptions with evidence.

Thank you for your time. I would welcome the chance to discuss how I can contribute to the team.

Sincerely,
Swati Aggrawal
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